

Soldiers are not allowed to march on a bridge.
This is to avoid resonant vibration of the bridge.
While crossing a bridge, if the period of stepping on the ground by marching soldiers equals the natural frequency of the bridge, it may result in resonance vibrations. This may be so large that the bridge may collapse.



Extra:

Pendulum in a lift:

(i) Lift moving upwards with acceleration a:

Effective acceleration due to gravity is $g_{eff} = g + a$

Then time period is $T = 2\pi\sqrt{\frac{l}{g_{eff}}} = 2\pi\sqrt{\frac{l}{(g+a)}}$

Since the time period is inversely related to acceleration due to gravity, time period will decrease when lift moves upward.

(ii) Lift moving downwards with acceleration a:

Effective acceleration due to gravity is $g_{eff} = g - a$

Then time period is $T = 2\pi\sqrt{\frac{l}{g_{eff}}} = 2\pi\sqrt{\frac{l}{(g-a)}}$

Since the time period is inversely related to acceleration due to gravity, time period will increase when lift moves downward.

(iii) Lift falls with acceleration $a > g$:

The effective acceleration is $g_{eff} = a - g$

Then time period is $T = 2\pi\sqrt{\frac{l}{g_{eff}}} = 2\pi\sqrt{\frac{l}{(a-g)}}$

in this case, the pendulum will turn upside down and will oscillate about higest point.

(iv) Lift falls with acceleration $a = g$:

The effective acceleration is $g_{eff} = g - g = 0$

Then time period is $T \to \infty$ which means pendulum does not oscillate and its motion is arrested.

(v) If the simple pendulum is kept in a car which moves horizontally with acceleration a:

The effective acceleration is $g_{eff} = \sqrt{g^2 + a^2}$

Time period is $T = 2\pi\sqrt{\frac{l}{g_{eff}}} = 2\pi\sqrt{\frac{l}{\sqrt{g^2 + a^2}}}$

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